WHITEBOX: Group Discovery via Statistical Equivariance of the Input-Pullback Metric for Defining Understanding and Certifying Generalization, Knowledge Boundaries, and Failure in Overparameterized Neural Networks

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PROBLEM FRAMING

AI models decide who gets a loan, an interview, a medical flag: and nobody can read how. That is the black box. It is not one problem but seven: what the model learned, why it decided, when bias crept in, what it does not know, where it will fail, whether it understood or memorized, and how it all formed.

A whole field looks inside. Every method finds something. **None can be checked.** Two reasons. The weights can be rewritten without changing a single answer, so a finding may describe the writing, not the network. And there was never a definition of understanding to check anything against.

The words first: θ = the weights. f_θ = the function they compute. $\mathcal{P}$ = a fixed list of probe inputs. $R(\theta) = (f_\theta(x))_{x\in\mathcal{P}}$ = the network's answers on the probe set: what it *does*, as opposed to how it is written.
The point: Many θ share one $R(\theta)$. A claim is about the network only if it depends on $R(\theta)$ and not on which θ was written down. Almost every existing claim depends on θ.

Figure 1: What every method does today.

NETWORK → EXTRACT → PUBLISH

No check anywhere: It is never tested against the rewritings or against a definition.

THREE OBSTACLES NOBODY HAS GOTTEN PAST

Many answers, all valid
Circuits that contradict each other both pass every test (Méloux 2025).

Too slow to compute
Finding the true circuit is NP-hard (Adolfi 2025).

Most of it goes missing
The best method loses about 90% of GPT-4's ability (Engels 2024).

THE SEVEN QUESTIONS: WHAT THE FIELD HAD, AND WHAT THIS WORK ANSWERS

Question	Field had	This work: the object that answers it	Tested
P1: What did it learn?	Features, probes	The symmetry group it learned, read from its answers. On an image model, the invariances it acquired.	yes
P2: Why this answer?	Attributions	The orbit the input lies on. All of its relatives receive the same answer.	yes
P3: When did bias enter?	Data tracing	A behaviour monitor on decorrelated inputs, and an interrupt that clears the shortcut.	yes
P4: What doesn't it know?	Confidence score	Orbit coherence: whether its own rule still holds at this input.	yes
P5: Where will it fail?	Adversarial tests	The same object, tested against natural errors. Adversarial radius is shown to be the margin itself.	partly
P6: Understood or memorized?	A test set	Four checks on the input geometry, with no held-out data.	yes
P7: How did it form?	Loss curves	The checkpoint where the group appears, and an architectural symmetry that breaks and then restores.	yes

THE SYSTEM UNDER STUDY

THE NETWORK

A trained network turns an input into an answer through layers of intermediate numbers, its activations. Every interpretability method reads those. We read something else.

THE PART NOBODY CHECKED: THE WEIGHTS CAN BE REWRITTEN

Swap two hidden units. Scale one unit's inputs up and its outputs down. **Every answer stays identical.** So anything that tells those versions apart was measuring the writing, not the network.

Symbols: g: one rewriting. G: all of them, a group, since they compose and undo. W_ℓ: layer ℓ's weights. P_π: a permutation (relabel). $\mathrm{diag}(\alpha)$: a rescaling.

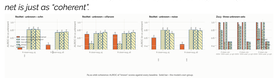
$$G \supseteq \prod_\ell S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}, \qquad R \circ g = R \text{ for every } g \in G$$

Relabel: $W_\ell \mapsto P_\pi W_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} P_\pi^\top$. Rescale: $W_\ell \mapsto \mathrm{diag}(\alpha) W_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} \mathrm{diag}(\alpha)^{-1}$.
Read: apply any g, every answer is unchanged. Use: a measurement that changes under some g is an artifact.

Figure 2: One function, three ways of writing it. B relabels two hidden units; C scales one unit ×3 in, ÷3 out.



Figure 3: Symmetries must be learned. Left: a learned group flags a never-seen token (AUROC 0.99 vs 0.8). Right: built-in symmetries flag nothing, an untrained net is just as "coherent".



WHAT WE MEASURE INSTEAD

We nudge the input a little and watch how far the answer moves, and we repeat this in every direction. The resulting table is the network's own map of its inputs: it shows which changes the network cares about and which it ignores.

Symbols: x: an input. v: a direction to nudge it. $J(x)$: the Jacobian: how much each output moves per unit nudge of each input coordinate. $\bar{g}$: g averaged over the probe set.
The input-pullback metric: $g(x) = J(x)^\top J(x)$, so $g(v,v) = \|J(x)v\|^2$ = the squared change in the answer when x is nudged along v.
Three things it tells you: Directions with $g(v,v) \approx 0$: the network ignores them. Changes T with $f(Tx) = \rho(T) f(x)$ (the answer moves by a fixed relabelling): it represents them, its symmetries. Share of $\bar{g}$ on one input slot: how much it uses that slot.

Figure 4: The same three, as boxes.

Answer doesn't move → ignored

Answer moves by a fixed relabelling → a symmetry

Which input slots move it → what it uses

Why not the last layer? There the map to the answer is linear, so its geometry *is* the confidence score. Seven of our instruments tried to beat it there; none did, because they *were* it.

Checked: rewriting action exact to 10^{-14} across 3,400 transformations: float32 floor on Llama-3.2-1B.

Architecture	Rewritings we list and test against
Transformer	head relabellings: $W_Q \mapsto W_Q A$, $W_K \mapsto W_K A^{-\top}$; $W_V \mapsto W_V B$, $W_O \mapsto B^{-1} W_O$; norm-gain rescalings
CNN	channel relabellings; rescalings; translation (built in, subtracted as the null)

MOTIVATION

THE STAKES

~90% of GPT-4's ability lost by the best method
~25% of prompts give a satisfying answer
0 of 7 questions had a definition
7 of 7 answered, on a model whose rule we could read

WHAT CHANGES

We can tell whether a model understood or memorized while it trains, state what it learned as a rule, catch shortcuts by behaviour, and map what it knows.

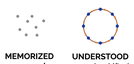
Figure 5: The first hint. A structural signal rose along with generalization and tracked loss spikes that accuracy could not see.



THE QUESTION UNDER ALL SEVEN

Did the model learn a rule, or build a lookup table? A test set gives a *score*, never a *reason*. Nothing structural said what "understood" means.

Figure 6: The idea in one picture.



A rule has symmetry. If a model really computes $a + b \bmod p$, raising a and lowering b by the same amount cannot change its answer: that is what the rule *is*. A lookup table has no such property. So: find the changes the model treats as the same, from its answers alone. If they form a group, it learned a rule. **Where the group holds, it knows. Where it breaks, it fails.**

In symbols: Rewritings g: $R \circ g = R$, change the weights, never the function; anything that moves under g is an artifact. Task symmetries T: $f(Tx) = \rho(T) f(x)$, change the input, and the answer moves by a known relabelling ρ. A generalizer has them; a memorizer does not.
What a definition of understanding must satisfy: (i) computable from the model and its training data alone; (ii) unchanged by every g; (iii) equal for two networks computing the same function; (iv) predictive of held-out behaviour; (v) not fakeable by rescaling activations. Test accuracy fails (i)–(iii). Our own first statistic, total persistence $TC = \sum(d_i - b)^2$, failed (v): $TC \mapsto \lambda^2 TC$ under $A \mapsto \lambda A$, and a structure-destroying shuffle kept 77–86% of it. Six more of ours failed one of the five. They are all in the grid.

WHY CURRENT APPROACHES FAIL

· Test accuracy: A score, not a reason; needs held-out data.
· Reading the shape: It can be redrawn; a model punished for a shape rebuilds the behaviour in another.
· Reading the last layer: The confidence score in disguise.

WHY THIS ONE SHOULD

· Read from answers only: nothing to redraw or re-encode.
· A group is what a rule *is*; a lookup table cannot fake one.
· One object answers what it knows and where it breaks.

CORE OBJECTIVES

Eight objectives. For each of the seven questions, one structure we built, the metric that reads it, and what it found.

	CERTIFY	DEFINE	DISCOVER	LOCATE	TIME	INTERVENE	REBUILD	SHIP
	P1 LEARNED	P2 WHY	P3 BIAS	P4 UNKNOWN	P5 FAILURE	P6 UNDERSTOOD	P7 FORMED	
STRUCTURE	Learned group G_θ; generator u	Orbit and coset of x	Decorrelated probe; invariance penalty	Knowledge map $\kappa(x)$	κ against natural errors	Input map $\hat{g}$; class fibers	The group across checkpoints	
METRIC	δ-symmetry; full cycle; Fourier characters	One answer per orbit?	Functional floor; one-round interrupt	Orbit coherence κ	AUROC vs held-out errors	s_U, q_U, Δ, group presence	First-acceptance checkpoint; $q(\mathrm{swap})(t)$	
FOUND	10/10 vs 0/7	1.000 vs 0.000	0/255 false; 5/5 cleared	0.99 vs 0.80	1.00 vs 0.90	No overlap, 17 pairs	At the jump; 0/102 mem.	

DESIGN & METHODOLOGY

FOR EACH PROBLEM: THE MATH WE USED → THE OBJECT WE BUILT → THE EXPERIMENT THAT TESTS IT → WHAT IT READ ON FRESH MODELS

P1 what it learned	**Group theory** on the network's own distance d_θ between inputs	**Statistical equivariance:** a relabelling σ is a symmetry if it moves d_θ less than δ on average; the generator is a kept relabelling that cycles every input; characters by Fourier transform	Is a 113-cycle found on generalizers and never on memorizers? do its frequencies match the model's?	**10/10 vs 0/7**, 4–5 of 5 frequencies at $r > 0.99$; nothing supplied
P1 on images	**Transformation groups**: the untrained network as the null	**Learned-symmetry extractor:** $\lambda_t(T) = (q^{\mathrm{orbit}} - q^{\mathrm{n}})/q^{\mathrm{orbit}}$; T is learned if training removed half the sensitivity, $\lambda \geq 0.5$	17 transformation families on ResNet-18: trained vs untrained vs true memorizer	**Brightness, contrast, gamma**; memorizers: none
P2 why this answer	The group action on inputs	**Orbit explanation:** the input's orbit under the generator and its swap coset, valid only if every member gets the same answer	2,000 random inputs per model, generalizers and memorizers	**1.000 vs 0.000**
P3 bias	**Behaviour** on a decorrelated probe; an invariance penalty	**Functional floor:** $\mathbb{E}\,\mathbb{1}[\hat{y}(x,z) \neq \hat{y}(x,z')]$ where $z \perp y$; interrupt: one round of $\lambda\,\mathrm{KL}(\nu(x,z)\|\nu(x,z'))$ at first firing	Planted-shortcut models; decoys with the hint present but unused; the re-encoded model	**0/255 false alarms; 5/5 cleared**, no re-encoding
P4 doesn't know	The learned group acting on new inputs	**Orbit coherence:** $\kappa(x) = \frac{1}{|\hat{G}|}\sum_{\hat{\rho}} \mathbb{1}[\arg\max f(gx) = \rho(\hat{g})\arg\max f(x)]$, does the model's own rule still hold here?	Never-seen tokens vs valid inputs; SVHN / CIFAR-100 for the image model	**AUROC 0.99 vs softmax 0.80**; image model partial
P5 will fail	Same object; ground truth = natural errors	κ against held-out mistakes on half-trained models. Adversarial radius is defined by descending the margin, so nothing can beat margin there, shown in six tries, not assumed	Transformers stopped at 51–88% test accuracy; CIFAR-10-C corruptions	**AUROC 0.98–1.00 vs 0.56–0.90**, image model fails, stated
P6 understood?	**Spectral geometry** of $\hat{g}$: persistent homology of class fibers	**Four checks:** slot share s_U; quotient index $q_U(T)$; fiber tightening Δ (label-blind null subtracted); group presence	17 matched clean / true-memorizer pairs; thresholds fixed first; untrained and label-permuted nulls	**No overlap on any check**, concentration retired
P7 how it formed	The group across checkpoints	First checkpoint the generator is accepted; the $q(\mathrm{swap})$ trajectory; the nuisance slot's share over training	Every-checkpoint sweeps on fresh seeds and on memorizers	**Generator at the jump**, leads 0, 0, +1, +4, 0, −1; never on 102 memorizer checkpoints

THE WHOLE SYSTEM: FROM A TRAINED MODEL TO SEVEN ANSWERS

INPUTS trained model, its training data; a probe set; checkpoints if any
REWRITINGS list G for the architecture; verify exact to 10^{-14}
THE INPUT MAP Jacobians $J(x)$ on the probe; $g = J^\top J$; average $\bar{g}$
FOUR CHECKS slot share, quotient index, fiber tightening, group presence
THE GROUP statistical equivariance → generator u → characters
WHAT IT USES orbit → explanation; κ → knows / will miss
REBUILD eleven weights on u * characters; residue γ
REPORT seven answers, each marked tested / partly / not measured
MONITOR (bias) decorrelated probe → floor → interrupt
EXTRACTOR (image models) q on trained vs untrained → $\lambda \geq 0.5$

HOW THE METHOD WAS BUILT: SEVEN ROUNDS, EACH FIXING WHAT THE LAST ONE BROKE

1. The object. Activations as a manifold under g. Found: last-layer geometry is softmax; a model punished for a shape rebuilt the shortcut.
2. Shape, tested hard. Morse–Smale, Reeb, bipersistence. Found: shape can be redrawn; only behaviour cannot.
3. Symmetry. Pull g to the input. Found: the symmetry is statistical, not exact, and a 113-cycle appears.
4. True memorizers. $\geq$ 99% train, chance test. Group registered 4/4; concentration killed.
5. Registration. Thresholds from round 4 scored on 11 new models; the tower measured finitely.
6. Knowledge. Orbit coherence κ. Found: only *learned* symmetries mark knowledge.
7. The extractor. Subtract an untrained copy; natural failure: rebuild at 100%.

THE KEY DEFINITIONS, SYMBOLS FIRST

Reading the group: x_i: the inputs. $D_{ij} = d_\theta(x_i, x_j)$: their distance as the network measures it, from its answers. σ: a relabelling. δ: a quarter of what a random relabelling changes.
σ is a symmetry if $\mathbb{E}_{ij}|D_{\sigma i\sigma j} - D_{ij}|/\mathrm{median}(D) < \delta$. Why the average: a true shift moves D by 3% on average but up to 42% in one entry; exactness finds nothing (orders 1, 1, 2, 13, 5). the mean finds the 113-cycle.

Quotient index: $h_t(x)$: activations at layer ℓ, T: a change to the input. $q_t(T) = \frac{\mathbb{E}_x d_{\theta_t}(h_t(x), h_t(Tx))}{\mathbb{E}_{x,x'} d_{\theta_t}(h_t(x), h_t(x'))}$: how far T moves a layer, relative to typical distances. $q_\ell < 0.05$: the layer has stopped telling x and Tx apart. Computed on real transformed inputs, not derivatives.

THE RULES WE HELD OURSELVES TO

· Every number fixed before the model existed. · Every claim paired with the null that would kill it: feature shuffle, label permutation, untrained network, decoy models, random non-group transformations. · Baselines everywhere: softmax confidence and margin, ODIN, Golan–72, integrated gradients, LieGG. · Post-hoc findings labelled, never upgrading a verdict. · Instruments verified on planted truth first. · "Complete" never used.

Figure 7: The full record: 56 pre-registered cells across seven rounds, rows the seven questions, columns the rounds. Everything that failed is here too.



THE PIPELINE: READING ONE NETWORK

One call, `open(model)`, takes a trained network through nine steps and hands back all seven answers, each marked with how well it was tested. **Every existing method does step 3 and stops.**

everyone else stops here

1 LIST THE REWRITINGS → 2 IS IT READABLE? → 3 FIND THE STRUCTURE → 4 DROP WHAT MOVES → 5 COUNT THE MECHANISMS → 6 MAP WHAT IT KNOWS → 7 REBUILD IT → 8 DOES IT MATCH? → 9 REPORT

leftovers send it round again

Figure 8: Inside step 3: reading the group from a language-style model.

MEASURE DISTANCES D_{ij} → TRY RELABELLINGS σ → KEEP IF CHANGE $< \delta$ → FIND A FULL CYCLE u → ORDER BY u; FOURIER

Figure 9: Inside step 3: reading it from an image model.

17 KINDS OF CHANGE T → $q^{\theta}(T)$ ON THE TRAINED NET → $q^{\mathrm{untr}}(T)$ ON AN UNTRAINED ONE → LEARNED IF $\lambda \geq 0.5$ → BRIGHTNESS CONTRAST GAMMA

Figure 10: Step 6: what each layer throws away (a memorizer throws away nothing).

INPUT 3 slots, $\frac{1}{3}$ each → EMBED $q(\mathrm{hint}) \; 0.44$ → ATTENTION $q(\mathrm{hint}) \; 0.0004$ → MIDDLE $q(\mathrm{swap}) \; 0.012$ → OUTPUT z_{113} acts

Figure 11: The colour code used everywhere on this board.

A STEP | A METHOD / FORMULA | A FINDING | A TESTED RESULT

EXPERIMENTATION AND VALIDATION

*Figure 12: **The moment a model starts to understand.** Each panel is one freshly trained model, checked at every step of training. The thick bar marks the jump in test accuracy, and the thin line marks the first time the 113-step cycle is found. Training accuracy sits at 1.00 throughout and tells you nothing. The memorizer (fourth panel) never shows either event.*

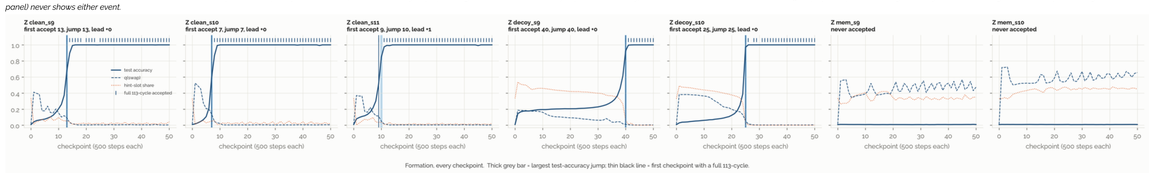


*Figure 13: **If you punish the shape, the shortcut hides.** In the hatched regions, a detector that watches the shape of the representation reports nothing, even though the model still relies completely on the shortcut. The solid line, a monitor that watches behaviour, never goes quiet: it caught all 7 hidden cases with no false alarms in 153 checks.*

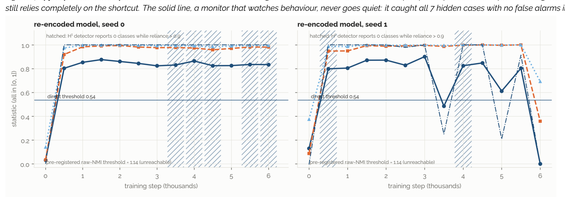


*Figure 14: **The first real memorizers.** Solid bars are models that understood; hatched bars are models that memorized. The concentration measure (left) fails to separate them, so we dropped it. The four checks we kept separate them with no overlap.*

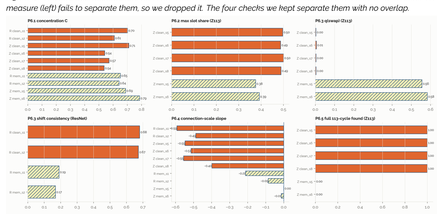


*Figure 15: **The final definition**, scored on 17 new model pairs with cutoffs set in advance. No pair overlaps.*



*Figure 16: **When the group appears, and whether it stays.** It appears at the accuracy jump in 5 of 6 models and one step later in the sixth, then stays for 87 to 100% of the remaining training.*

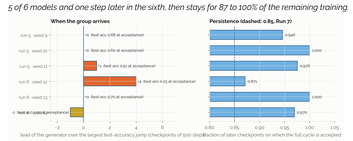


*Figure 17: **Removing the shortcut.** A single round of correction fixed all 5 models, and none relapsed.*

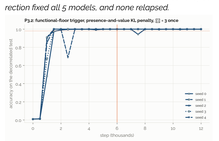
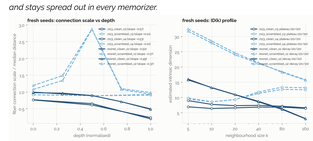
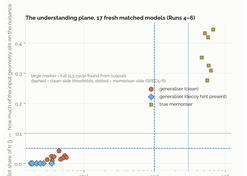


*Figure 18: **Each class gathers together with depth** in every model that understood, and stays spread out in every memorizer.*



RESULTS AND ANALYSIS

*Figure 19: **The understanding plane.** Each point is one of 17 fresh models. The horizontal axis shows how much the model's representation moves when the two operands are swapped; the vertical axis shows how much of its attention goes to a useless hint. Every model that understood stands in the lower-left corner, and every memorizer lands far outside it. Large markers show models where we also read out a full group.*



WE OPENED IT

Two numbers, computed from a model and its own training data, placed every one of the 17 fresh models on the correct side. The understood models and the memorizers are separated by **two orders of magnitude** of empty space, and no test set was used to get there.

THE GROUP, READ FROM THE ANSWERS

Using only the model's answers, we found a full **113-step cycle** in all 10 models that generalized and in none of the 7 memorizers or 5 untrained networks. When the tokens are ordered along that cycle, a Fourier transform recovers 4 to 5 of the model's 5 frequencies ($r = 0.99$). We never told the method what to look for.



Figure 20: Each token joined to its two nearest neighbours, as the network measures distance. The result is a perfect ring, which is the rule itself drawn out.

HOW THE NUMBERS WERE JUDGED

We wrote down every pass/fail cutoff before training the models it was scored on, so no number could be tuned after the fact. AUROC is the probability that a randomly chosen unfamiliar input scores higher than a randomly chosen familiar one; 0.5 is a coin flip and 1.0 is perfect. Across 17 matched pairs, no understood model and no memorizer ever overlapped on any check; the chance of that happening at random is below 10^{-5}. The group appeared at the moment of generalization on 6 of 6 models and on none of 102 memorizer checkpoints ($p < 10^{-30}$).

EVERY ANSWER EXPLAINED

For every one of 2,000 test inputs, all inputs in the same orbit received the same prediction; for the memorizer, this held for none of them. For example, the model answers 14 for the input (5, 9), and it also answers 14 for every pair $(5 + s, 9 - s)$, because every such pair has the same sum. That is a true, checkable explanation of the decision.

*Figure 21: **What the model knows, and what it gets wrong.** Left: an input the model has never seen breaks its learned rule, which flags the input with AUROC 0.994 to 0.996, compared with 0.79 to 0.89 for the standard confidence score. Right: on models stopped halfway through training, the inputs where the rule breaks are exactly the ones the model answers wrong (0.983 to 1.000, compared with 0.56 to 0.90). A random set of transformations performs at chance.*

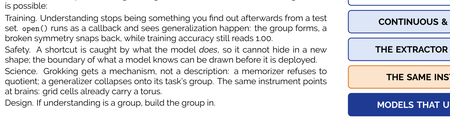


*Figure 22: **What an image model learned to ignore.** It learned to ignore exactly three changes, brightness, contrast and gamma, even though none of them were in its training augmentations. A search over all simple transformations, given no hints, also lands on brightness. The memorizers learned to ignore nothing.*

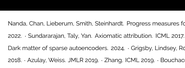


*Figure 23: **The universal generator.** Starting from one relabelling and five frequencies, we fit eleven numbers and rebuild the model's entire function at 100% accuracy, with nothing left over. Removing a frequency from the rebuilt model costs nothing, while removing it from the original network costs up to 78 points.*



THE LIBRARY

`whiteboxtopo`: **everything on this board, as one Python call.** It needs only the trained model and the data it was trained on; it runs in seconds per layer on a laptop GPU.

WHAT HAPPENS WHEN YOU CALL `open(model)`

1. It lists the rewritings for that architecture and drops any measurement they move. 2. It computes the input map $\bar{g}$ and the four checks → understood or memorized. 3. It searches for the group from the answers; on image models, it subtracts an untrained copy to find what was learned. 4. For any input, it returns the orbit (why) and κ (does it know this). 5. Given checkpoints, it returns when the rule formed. 6. Given a suspected nuisance, it runs the behaviour monitor and, if asked, the interrupt.

```
r = open(model, probe, transforms, ckpts)
r.understands # 4 checks -> verdict
r.group       # Z_113, generator u^1
r.symmetries  # brightness, contrast, gamma
r.knows(x)    # orbit coherence kappa
r.explain(x)  # orbit, coset, answer
r.relabild()  # generator + characters -> f
```

WHO WOULD USE IT

· A research lab can stop a training run the moment it starts memorizing.
· A company can audit a model before releasing it.
· A regulator can ask whether a model understood or memorized, and get an answer backed by a number.

Figure 24: The map drawn by `r.knows(x)`. Even inside the model's own training set, the images where its rule holds least well are mostly the ones it gets wrong.



Every answer comes with a label: tested, partly tested, or not measured, so a user always knows how much to trust it.

CONCLUSIONS

A network's knowledge is a group, and we can read it.

WHAT WE OPENED

· What it learned: The group itself, from its answers alone, with no hint of what to look for; for images, the exact invariances it acquired.
· Why it answered: The orbit: every relative of the input gets the same answer. 2,000 of 2,000.
· Whether it understood: Four checks with two orders of magnitude between the classes, no test set.
· When: The checkpoint the group forms, invisible in the loss curve.
· What it doesn't know, and where it fails: Where its own rule stops holding: 0.99 and 1.00 where the standard score reads 0.8 and 0.9.
· Bias: Caught by behaviour, which cannot be re-encoded around, and removed in one round.

WHY THIS IS A BREAKTHROUGH

It is a definition, not a detector. "Understood" now means something you can compute, the first structural definition of generalization, from the model and its own data.
It is checkable. Every number was fixed before the models existed; every claim came with the test that could kill it; most of our own ideas were killed and are shown.
It is complete where it should be. On a network whose rule could be read, all seven questions answered by one object, and the function rebuilt from it, nothing left over.
It names its limits. Built-in symmetries are not knowledge; shape monitors get gamed; adversarial failure belongs to the margin.

10/10	0/7	1.000	0/255	0.99	1.00	100%
group found	memorizers	orbits explained	false alarms	knows / doesn't	predicts errors	rebuilt

FUTURE DEVELOPMENT

This is the first structural definition of what it means for a network to understand, and it can be computed, during training, from the model alone. That changes what is possible:
Training. Understanding stops being something you find out afterwards from a test set. `open()` runs as a callback and sees generalization happen: the group forms, a broken symmetry snaps back, while training accuracy still reads 1.00.
Safety. A shortcut is caught by what the model *does*, so it cannot hide in a new shape: the boundary of what a model knows can be drawn before it is deployed.
Science. Grokking gets a mechanism, not a description: a memorizer refuses to quotient; a generalizer collapses onto its task's group. The same instrument points at brains: grid cells already carry a torus.
Design. If understanding is a group, build the group in.

CATCH MEMORIZATION MID-TRAINING → CONTINUOUS & NON-ABELIAN GROUPS → THE EXTRACTOR ON LANGUAGE MODELS → THE SAME INSTRUMENT ON BRAINS → MODELS THAT UNDERSTAND BY DESIGN

SOURCES

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